JUNGHYUN (JOANNE) YUN

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RESEARCH INTERESTS

Multimodal Data Analysis, Machine Learning, In-Situ Process Monitoring, Federated Learning

EDUCATION

University of Wisconsin–Madison — Madison, WI
M.S. in Industrial & Systems Engineering — *Sep 2025 – Present*

- Advisors: Prof. Hantang Qin and Prof. Kaibo Liu
- GPA: 3.9/4.0
- Coursework: Human Factors of Data Science & Machine Learning, Signals and Systems; (as Visiting Student) Industrial Data Analysis, Simulation & Probabilistic Modeling

Ajou University — Suwon, South Korea
B.S. in Industrial Engineering — *Mar 2023 – Aug 2026*

- Advisor: Prof. Jooyeon Lee
- GPA: 4.26/4.5 · **Class rank: 1/139**
- Coursework: Quality Engineering, Reliability Engineering, Manufacturing Engineering, IE Machine Learning

PUBLICATIONS

[1] P. Zhang, **J. Yun**, and H. Qin. "Federated Learning-Based Regression Framework to Predict Printed Line Width in Electrohydrodynamic Inkjet Printing." *NAMRC 54*, June 2026. *(Accepted)*

[2] **J. Yun**, H. Ouidadi, K. Alharbi, P. Zhang, and H. Qin. "End-to-End Computer Vision Pipeline for Perforation Defect Detection and Classification in Industrial Imaging." *IEOM Society International*, June 2026. *(Accepted)*

[3] H. Ouidadi, K. Alharbi, P. Zhang, **J. Yun**, and H. Qin. "Deep Learning-Based Perforation Defect Detection and Classification for Automated Industrial Inspection: An Industry Case Study." *(In preparation)*

RESEARCH EXPERIENCE

FEAP Lab, University of Wisconsin–Madison — Madison, WI
Team Lead | Multimodal Process Monitoring for Metal AM — *Sep 2026 – Present*

- Lead the FIRE MBSE TG3 acoustic team; built an **acoustic and video monitoring pipeline** for LPBF to predict per-sample ductility and yield strength, supporting the **DOE FIRE project**
- Segmented 209 layers from 5 hr build audio using signal processing (acoustic equalization) and NCC template matching, replacing earlier RMS and watchdog approaches that produced 415–3,039 false layer boundaries
- Designed a video laser-off detector using keyframe sampling with Otsu-derived thresholds, recovering **95 laser-off intervals** from ∼620k frames in ∼2.5 min on a laptop
- Developing a homography-based part map that labels all 353 build-plate samples in chamber-camera frames (331 of 346 pooled detections matched to the reference grid)

LAMM Lab, Purdue University — West Lafayette, IN
Research Intern (Advisor: Prof. Martin Jun) — *Jun 2026 – Aug 2026*

Acoustic Process Monitoring for Directed Energy Deposition

- Synchronized audio, video, and machine logs for a DED isolation experiment (powder, stage, and laser switched independently) by regressing panel O_2 readings against the log (residual 0.17 s), correcting a manual offset estimate by 3.4 s

- Showed laser on/off is undetectable from band energy ($<$0.4 dB), then recovered a laser-linked **7–8.5 kHz burst** via event-related averaging over 45 laser-on edges ($\approx 6\sigma$), absent in the laser-off control run

Targetless Surround-View Camera Calibration (with Miratek and Hyundai Translead)

- Proposed a target-free extrinsic calibration method for commercial-vehicle fisheye cameras that uses the vehicle body boundary as a geometric constraint, estimating roll $\rightarrow$ yaw with gradient-based boundary detection and Huber-loss optimization
- Achieved a **median roll error of 1.21–1.68**° across 232 camera views from 58 real bus images, and cut worst-case error from 26.8° to 19.2° versus the existing method in a C++/OpenCV evaluation harness
- Identified a per-camera systematic bias behind about half of the remaining error, and a simulator update that silently corrupted ground truth for 56 of 58 sessions

FEAP Lab, University of Wisconsin–Madison Madison, WI
Research Assistant (Advisor: Prof. Hantang Qin) *Sep 2025 – May 2026*

Federated Learning for Electrohydrodynamic (EHD) Inkjet Printing

- Built per-client XGBoost regressors for **8 EHD printing systems** (1,185 runs) as the individual-learning baseline of a federated learning framework for line-width prediction
- Showed that the best single-client model (R^2 0.983 within its own client) **falls below zero R^2 on 4 of 8 clients**, the evidence that motivates federated training
- Traced the failure to non-IID client data (different substrates, printers, operators, and feature availability) and built the pandas ETL pipeline that made client datasets comparable

Industrial Surface Defect Detection

- Co-developed an adaptive-thresholding pseudo-labeler that achieved **93.4% region localization**
- Designed a label-efficient sampling strategy that selected **900 of 25,000** unlabeled production images
- Trained a Faster R-CNN detector (AP50 96.8%), placing **3rd of 33 teams** at the IISE QCRE Data Challenge 2026; later rebuilt the lost pipeline in Detectron2 and reproduced it at AP 85.0 / AP50 97.6
- Ran a component ablation (213 images, 852 regions) and a 6-parameter sensitivity analysis, showing localization stays at 92–95% across all settings
- Designed a center-based evaluation metric after finding that over-padded ground-truth boxes make IoU@0.5 penalize accurate detections
- Co-writing the methods section (adaptive thresholding) of a journal manuscript in preparation

New Industrial Convergence R&D Lab, Ajou University Suwon, South Korea
Undergraduate Research Intern (Advisor: Prof. Jooyeon Lee) *Mar 2023 – Aug 2025*

- Developed an anomaly detection model on module-level power electronics (MLPE) telemetry for solar panel energy management, in a funded project with Korea Midland Power (KOMIPO) and Kernelog
- Built API-based data acquisition pipelines for digital-twin monitoring of solar panels
- Led a microfluidic sweat-extraction patch project with colorimetric, image-based pH sensing (2nd Place, Field-based Technical Problem Solving Project 2024)
- Presented "Bicycle Lift Installation Optimization based on Spatial Information" (oral), The Korean Society of Systems Engineering, May 2025

Honors & Awards

IEOM Senior Capstone Design Competition 2026, 2nd Place	2026
IISE QCRE Data Challenge 2026, 3rd Place (of 33 teams)	May 2026
UW–Madison LikeLion Hackathon 2026, 1st Place	Apr 2026
Cursor Hackathon 2026, Finalist	Apr 2026
UW–Madison College of Engineering Makerthon, 4th Place	Sep 2025
Field-based Technical Problem Solving Project, MOTIR, South Korea, 2nd Place	Nov 2024

Public Data Analysis Idea Contest, Seongnam City, South Korea, 3rd Place — Jul 2024
Guwon Scholarship, South Korea (Top 4%) — Sep 2023 – Jul 2024
Ajou University Scholarships (≈$20,000 total) — Mar 2023 – Sep 2025
10th Ajou Hidden Champion Contest, 1st Place — Aug 2023

Entrepreneurship

NSF I-Corps Quantum+X — Madison, WI
Co-Technical Lead — *Sep 2026 – Oct 2026*

- Interviewed quantum-industry leaders (Quantinuum, Microsoft Quantum, IonQ, Fujitsu Research, and others) on how quantum technology could support a RAG-based LLM for fusion research

NSF I-Corps BOOST, Great Lakes Regional Hub — Madison, WI
Co-Entrepreneurial Lead — *Jun 2026 – Aug 2026*

- Led customer discovery with nuclear-industry practitioners on how operational knowledge is captured and transferred, refining a value proposition for a source-grounded AI assistant
- Built a Google Apps Script booking app so interviewees could self-schedule across time zones

In total, 50+ customer-discovery interviews across two NSF I-Corps programs.

Teaching Experience

University of Wisconsin–Madison — Madison, WI
Reader/Grader, ISyE 320 Simulation and Probabilistic Modeling (Prof. Qiaomin Xie) — *Sep 2026 – Present*

- Sole grader for 80+ students; graded homework and provided feedback

Purdue University — West Lafayette, IN
Teaching Assistant, KIAT Data Analytics Program (Prof. Martin Jun) — *Jun 2026 – Aug 2026*

- Ran hands-on Python labs and debugged student code for 90+ students across 20 sessions (∼60 contact hours), with bilingual English/Korean support

Ajou University — Suwon, South Korea
Teaching Assistant, Future Industrial Revolution and Technology Start-Up (Prof. Jinam Wang) — *Jul 2025*

- Organized the final presentation event for 50+ students and facilitated team project sessions

Hyundai Motor Group — Seoul, South Korea
Education Volunteer, H-Jump School Education Volunteer Program — *Mar 2024 – Nov 2024*

- Mentored high school students and taught English and math (100+ hours)

Project Experience

Polaris: MindGraph | *Finalist, Cursor Hackathon 2026* — *Mar 2026 – Apr 2026*

- Built a research tool that generates literature-grounded knowledge graphs from OpenAlex retrieval and Gemini extraction (React, TypeScript, Express)

Incident Copilot | *1st Place, UW–Madison LikeLion Hackathon 2026* — *Apr 2026*

- Built an AI incident-response console with staged inference (triage, root cause, action plan) and human-in-the-loop safety gates in one day (Next.js, FastAPI)

OraQ: AI-Based Oral Cancer Screening | *LikeLion US Data Science Team* — *Jan 2026 – Apr 2026*

- Prepared the oral-image dataset and built the full FastAPI backend (Google/JWT authentication, inference pipeline, scan history) for a mobile screening service

Raspberry Pi PCB Defect Detector | *4th Place, Plexus Challenge, UW–Madison Makerthon* — *Sep 2025*

- Proposed a component-reference matching approach for detecting missing or misplaced parts, built as a Raspberry Pi computer-vision prototype with a team of three

3D Virtual Aircraft Assembly Simulator | *Advisor: Prof. Jeongsam Yang* — *Mar 2025 – Jul 2025*

- Built an interactive jet-engine assembly and disassembly training environment in Unity (C#)

Resume-Based Career Guidance Model | *IE Machine Learning course* *Mar 2025 – Jul 2025*

- Predicted career pathways of IE students with decision tree, LDA, and logistic regression, handling class imbalance with SMOTE and class weights

Aircraft Engine Fault Detection | *Quality Engineering course, PHM 2024 data* *Sep 2024 – Dec 2024*

- Detected engine faults with $\bar{X}$-R and $\bar{X}$-S control charts and a random forest model

Leadership & Service

LikeLion US, University of Wisconsin–Madison Madison, WI
Public Relations Manager; Data Science Team Member *Jan 2026 – Present*

McBurney Disability Resource Center, University of Wisconsin–Madison Madison, WI
Notetaker *Sep 2025 – May 2026*

- Produced structured lecture notes for students with accommodations

WAI Career Organization, Ajou University Industrial Engineering Suwon, South Korea
Vice President (Mar 2024 – Aug 2024); Treasurer (Sep 2023 – Feb 2024) *Sep 2023 – Aug 2024*

- Organized career talks with industry professionals and alumni for 100+ students; managed the organization's budget as Treasurer

Technical Skills

Programming	Python, R, C++, MATLAB, C#
Frameworks	PyTorch, TensorFlow, Detectron2, OpenCV, scikit-learn, pandas, Matplotlib
Machine Learning	CNNs (YOLO, Faster R-CNN), RNNs (RNN, LSTM), Encoder–Decoder models, SVM, XGBoost
Signal Processing	Fast Fourier Transform, STFT/spectrograms, acoustic equalization, template matching (NCC)
Tools	Git, CMake, FastAPI, QGIS, Unity